MGMTMSA-400-01/02
Statistical Foundations for Data Analytics

Instructor: Professor Yihao Yuan (yihao.yuan@anderson.ucla.edu)
Office Hours: Wednesday 4–5pm, B516 (Gold Hall)

Teaching Assistant: Charlotte Zhu (charlotte.zhu.phd@anderson.ucla.edu)

Class Meeting Times

Section 1: Tuesday, 1:00–3:50pm, D301
Section 2: Tuesday, 4:10–7:00pm, B313

We will have eight regular lectures on the following dates: 9/22, 9/29, 10/6, 10/20, 11/3, 11/10, 11/17, and 12/1. Midterm will be held on 10/27, and final on 12/8. All these dates are Tuesdays.

Course Overview

This course introduces the statistical foundations of data analytics. Key topics include statistical inference, linear regression, and logistic regression. The goal is to provide students with a solid grounding in core techniques and to prepare students for advanced coursework in analytics and machine learning.

Course Policies and Requirements

Prerequisites. You are expected to be familiar with exponential and logarithmic functions, summation notation, and basic linear algebra (vectors, matrices, and matrix multiplication).

Required Software. R (see Software section below).

Textbook. There is no required textbook. There are four optional textbooks:

1. *Statistics: The Art and Science of Learning from Data* by Agresti, Franklin, and Klingenberg
2. *R for Everyone: Advanced Analytics and Graphics* by Lander
3. *Business Data Science* by Taddy
4. *Mastering Metrics: The Path from Cause to Effect* by Angrist and Pischke

Bruin Learn. All announcements, readings, lecture slides, and recordings will be posted on Bruin Learn. Please check them regularly.

Lecture Slides. Slides will be posted before lectures whenever possible. However, students are strongly encouraged to take their own notes and engage actively during class.

Lecture Recordings. All lectures will be recorded. Recordings are not available by default; attendance in person is expected. If you must miss a lecture and would like access to the corresponding recording, please email the instructor.

Grades. Grades will be posted on Bruin Learn and Gradescope once released. Please do not email the instructor or TA for updates.

Assignments. All assignments must be submitted via Gradescope (accessible through Bruin Learn). No late submissions will be accepted, and no make-up assignments are offered.

iClicker. We will use iClicker to track attendance. You must create an iClicker account, and use that account when you attend, for your attendance to be recorded.

TA Review Sessions. The TA will hold four review sessions **on Zoom**. The first session will provide an introduction to using R. The remaining sessions will review problem sets. You are encouraged to bring any questions about the problem sets or course materials. TA sessions will be offered on the following dates:

- Session 1: Thursday, 10/1, 4–5pm
- Session 2: Thursday, 10/22, 4–5pm
- Session 3: Thursday, 11/19, 4–5pm
- Session 4: Friday, 12/4, 4:30–6:00pm

Ask Questions. If you have questions about the course material, please attend the instructor's office hours or TA's review sessions, or email the TA for help. If the TA is unable to resolve your question, you may then email the instructor.

Use of AI. You may use generative AI tools (e.g., ChatGPT) to help complete assignments (except exams) at your own discretion. Generative tools can be unreliable; you are responsible for any work derived from them.

Software

All computing will be done with R. R is a free programming environment and language for statistics and data analytics. We will use only a subset of R's functionality, and all necessary material will be covered in class. Prior familiarity with R is not required. Use of RStudio IDE (integrated development environment), a user-friendly interface for R, is encouraged. Helpful R resources and videos will be posted on Bruin Learn.

Analyses will be demonstrated in class using R. Supplementary Python code reproducing course examples will also be provided, though Python will not be illustrated or covered in lectures.

Homework Software Policy: You may use any software (e.g., R, Excel, Python, Stata, Julia) for homework, but support will only be provided for R.

Deliverables and Grading

- 20% — Mid-quarter exam
- 35% — Final exam
- 40% — Assignments (4 assignments, 10% each)
- 5% — Attendance

Exams

Two joint exams (for both sections) will test comprehension of course concepts:

- **Mid-quarter exam**: 10/27, 1–2:30pm, Crown Auditorium
- **Final exam**: 12/8, 1–2:30pm, Crown Auditorium

Format. Both exams are multiple choice, with all questions weighted equally. Unless otherwise noted, each question has exactly one correct answer. Expect 27–30 questions, and 90 minutes to complete the exam.

Practice exams. Exams and practice tests from previous years, along with their solutions, will be distributed on BruinLearn in advance to help you prepare.

Calculators. You are expected to bring your own scientific calculator to each exam. I will bring five spare calculators in case of malfunction, but these are available on a first-come, first-served basis; **once they are gone, no additional backups will be provided, and you are responsible for arriving prepared**. More advanced calculators are permitted, though any functionality beyond that of a standard scientific calculator will be of no use on the exam.

Cheat Sheet. Both exams are **closed-book** and **closed-notes**. However, you may bring one 8.5" $\times$ 11" cheat sheet to an exam. You may write on both sides of your cheat sheet. You may not bring both your midterm cheat sheet and a new cheat sheet to the final exam. Put your name on your cheat sheet and turn in your cheat sheet with your exam. Two sheets of paper with only notation on one side DO NOT constitute one cheat sheet. A piece of paper with multiple pages attached to it does NOT constitute one cheat sheet. Any student who violates the cheat sheet policy will receive a penalty of 10% of their unpenalized exam score (i.e. only receive credit for 90% of the exam). The cheatsheet for the midterm will be returned to you during the next lecture from the midterm exam.

Rescheduling. I will only entertain requests for rescheduling an exam in cases of **extreme** medical or personal emergency, and any request must be made **in advance**. Emergencies do not include job interviews, weddings, vacations, class trips, or having one or more other exams or assignments due on the same day. Plan ahead and manage your time.

Students with Disabilities. If you have a learning disability and require special provisions for taking exams, it is **your responsibility** to get in touch with UCLA CAE (https://www.cae.ucla.edu) and to have them contact me directly to make exam arrangements for you. I will email the exam to them, and they will administer it.

Re-Grade Requests. Students may request a re-grade for an exam within 14 days of the exam. Clerical errors will be corrected at no risk to the student. All other re-grading requests will result in a re-grade of the entire exam. Downward as well as upward grade revisions are possible. To request a re-grade, the student must provide a written request for a regrade along with the exam. **The request must be sent to the instructor**. The request letter must clearly state the basis for the re-grade request.

Assignments

Four assignments focus on applying course ideas and methods. You may use any tool; however, we provide support only for R.

Students may work individually or in groups of up to four. If you work as a group, all group members much come from the same section. You do not need to keep the same group for all assignments. Enter group members when submitting on Gradescope. Only one submission per group is required on Gradescope. Individual submissions from group members are not necessary. **Late assignments will not be accepted. If a group submits late or fails to submit, all members of that group will receive a grade of zero for that assignment.**

Assignment Schedule. The timeline for the four assignments is as follows:

- Assignment 1: available Wednesday, 9/23; due Sunday, 10/11, at 11:59pm
- Assignment 2: available Wednesday, 10/7; due Sunday, 10/25, at 11:59pm
- Assignment 3: available Wednesday, 11/4; due Sunday, 11/22, at 11:59pm
- Assignment 4: available Wednesday, 11/18; due Friday, 12/4, at 11:59pm

Attendance

You must attend the section in which you are registered. **iClicker will be used to track attendance separately for each section.** Attendance will be checked from the second lecture. You may miss up to two lectures without penalty; beyond that, each missed poll will deduct 1 point from your final grade.

Final Course Letter Grades

Letter grades include A+, A, A−, B+, B, B−, C, and F. Roughly 30–40% of students will receive an A or A+, and a small number will receive a B− or below. For students within 1% of a cutoff (the "rounding distance"), I will consider a bump to the next letter grade only if you have participated actively and thoughtfully in class. **Any such adjustment is entirely at my discretion and is not subject to request or negotiation.** There are no extra credit assignments.

Course Outline

We will cover the following eight modules over the quarter. A module may take more or less than one week.

1. **Course Overview and Introduction to R**
   - Focus and Scope of the Course
   - A Short Introduction to R
2. **Review of Statistics**
   - The Normal Mean Problem
   - Estimation of the Mean and Variance
   - Confidence Intervals and Significance
   - Central Limit Theorem
   - Hypothesis Testing
3. **Introduction to Regression**
   - Conditional Prediction
   - Linear Prediction and Fitting
   - Scoring Model Fit and Least Squares
   - Intuition behind the Least Squares Solutions
   - A Goodness of Fit Measure: R^2
4. **Estimation, Testing, and Prediction in the Simple Linear Regression Model**
   - Prediction under the Simple Linear Regression Model
   - The Error Term and the Conditional Distribution of Y on X
   - Statistical Inference on the Estimates
   - Construction of Prediction Intervals
5. **Extension of Simple Linear Regression**
   - Log-Log Regressions
   - Slicing and Smoothing
   - Heteroscedasticity
6. **The Multiple Regression Model**
   - Model Setup and Roadmap
   - Estimation
   - Inference and Prediction
   - Too Many Input Variables: Regularization
   - Too Few Input Variables: Omitted Variables, Their Impact, and Fixes
   - Interaction Effect
7. **Logistic Regression**
   - The Setup of the Logistic Regression Model
   - Interpretation of Logistic Regression Slope Coefficients
   - Estimation: The Likelihood Funcftion
   - Inference and Prediction
8. **Multinomial Logit Model**
   - The Setup, Estimation, and Interpretation
   - Multinomial Logit Model with Aggregate Data